

- 2** Four students **A**, **B**, **C** and **D** measured the potential difference across an electric component. Each student obtained five sets of data for the potential difference.
- If the actual potential difference is $1.65 \times 10^{-2} \text{ V}$, which student's measurement was accurate but not precise?

	Potential difference / 10^{-2} V					Mean / 10^{-2} V
A	1.61	1.65	1.65	1.58	1.61	1.62
B	1.66	1.68	1.63	1.64	1.65	1.65
C	1.65	1.64	1.65	1.65	1.66	1.65
D	1.69	1.70	1.69	1.70	1.69	1.69

Answer: B

Student A's results: mean value = $1.62 \times 10^{-2} \text{ V}$, spread = $0.07 \times 10^{-2} \text{ V}$

Student B's results: mean value = $1.65 \times 10^{-2} \text{ V}$, spread = $0.05 \times 10^{-2} \text{ V}$

Student C's results: mean value = $1.65 \times 10^{-2} \text{ V}$, spread = $0.02 \times 10^{-2} \text{ V}$

Student D's results: mean value = $1.69 \times 10^{-2} \text{ V}$, spread = $0.01 \times 10^{-2} \text{ V}$

The actual potential difference is $1.65 \times 10^{-2} \text{ V}$

Hence, student B's results are accurate (low systematic errors, mean value close to actual value) but not precise (high random errors, large spread).

- 3 A boat moves at velocity of 4 m s^{-1} due south in still water across a lake. It then experiences